



Actividad: Diferenciación e integración de series de Potencias.

1) Calcular usando serie geometrica

$$\left(\sum_{n=0}^{\infty} x^n \right)''$$

$$\sum_{n=0}^{\infty} n(n-1)x^{n-2} = \frac{2}{(1-x)^3}$$

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x} \quad \text{para } |x| < 1$$

$$\left(\sum_{n=0}^{\infty} x^n \right)'' = \sum_{n=2}^{\infty} n(n-1)x^{n-2} \quad \text{Radio } (-1; 1)$$

$$\left(\frac{1}{1-x} \right)'' = \left((1-x)^{-1} \right)'' = \left((-1)(1-x)^{-2}(-1) \right)'' = \left((1-x)^{-2} \right)''$$

$$= (-2)(1-x)^{-3}(-1) = \frac{2}{(1-x)^3}$$

$$\sum_{n=2}^{\infty} n(n-1)x^{n-2} = \frac{2}{(1-x)^3} \quad \text{para } |x| < 1$$

2) Demuestra que $(\sin(x))' = \cos(x)$ y $(\cos(x))' = -\sin(x)$
 $\forall x \in \mathbb{R}$.

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!} = \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1)x^{(2n+1)-1}}{(2n+1)!}$$

$$(\sin(x))' = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)} = \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n+1}}{(2n+1)!}$$

$$(\sin(x))' = \sum_{n=0}^{\infty} \frac{(-1)^n (2n+1) x^{2n}}{(2n+1)(2n)!} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$(\sin(x))' = \cos(x) \quad \forall x \in \mathbb{R}$$

b) $(\cos(x))' = -\sin(x)$

$$\sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$\cos(x) = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n}}{(2n)!}$$

$$(\cos(x))' = \sum_{n=1}^{\infty} \frac{(-1)^n (2n) x^{2n-1}}{(2n-1)! (2n)} = \sum_{n=1}^{\infty} \frac{(-1)^n x^{2n-1}}{(2n-1)!}$$

$$(\cos(x))' = \sum_{k=0}^{\infty} \frac{(-1)^{k+1} x^{2(k+1)-1}}{(2(k+1)-1)!} = \sum_{k=0}^{\infty} \frac{-1(-1)^k x^{2k+1}}{(2k+1)!}$$

$$(\cos(x))' = - \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

$$(\cos(x))' = -\sin(x) \quad \forall x \in \mathbb{R}$$

3) Derivar $f(x)$ by using serie geométrica

$$f(x) = \arctan(x) \quad f'(x) = \frac{1}{1+x^2}$$

$$\sum_{n=0}^{\infty} \frac{1}{1-y}$$

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$$m = -2 \quad \frac{1}{1 - (-x^2)} = \sum_{n=0}^{\infty} (-x^2)^n = \sum_{n=0}^{\infty} (-1)^n x^{2n}$$

$$|1 - x^2| < 1 \Rightarrow |x| < 1$$

$$\text{Radio} = 1$$

$$\int_0^1 f'(t) dt = \int_0^x \left(\sum_{n=0}^{\infty} (-1)^n t^{2n} \right) dt$$

$$f(x) - f(0) = \sum_{n=0}^{\infty} (-1)^n \int_0^x t^{2n} dt$$

$$\text{Radio} = 1$$

$$\arctan(x) - 0 = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{2n+1}$$

4) Calcular usando serie de Maclaurin e^x

$$\int_a^b e^{x^2} dx$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$e^{x^2} = \sum_{n=0}^{\infty} \frac{(x^2)^n}{n!} = \sum_{n=0}^{\infty} \frac{x^{2n}}{n!}$$

$$\int_a^b e^{x^2} dx = \int_a^b \left(\sum_{n=0}^{\infty} \frac{x^{2n}}{n!} \right) dx = \sum_{n=0}^{\infty} \frac{1}{n!} \int_a^b x^{2n} dx$$

$$\int_a^b e^{x^2} dx = \sum_{n=0}^{\infty} \frac{1}{n!} \left[\frac{x^{2n+1}}{2n+1} \right]_a^b = \sum_{n=0}^{\infty} \frac{1}{n!(2n+1)} \left(b^{2n+1} - a^{2n+1} \right)$$

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5) Calcular usando la serie de Maclaurin de $\sin(x)$.

$$\int_a^b \frac{\sin(x^2)}{x} dx \quad \sin(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{4n+2}$$

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1}$$

$$\sin(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{4n+2}$$

$$\int_a^b \frac{\sin(x^2)}{x} dx = \int_a^b \left(\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{4n+1} \right) dx$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \int_a^b x^{4n+1} dx = \int_a^b \frac{\sin(x^2)}{x} dx \left[\frac{x^{4n+2}}{4n+2} \right]_a^b$$

$$\sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)(4n+2)} (b^{4n+2} - a^{4n+2})$$